

## Section 7: Radioactivity and particles

- a) Units
- b) Radioactivity
- c) Particles

### a) Units

*Students will be assessed on their ability to:*

- 7.1 use the following units: becquerel (Bq), centimetre (cm), hour (h), minute (min), second (s).

### b) Radioactivity

*Students will be assessed on their ability to:*

- 7.2 describe the structure of an atom in terms of protons, neutrons and electrons and use symbols such as  $^{14}_6\text{C}$  to describe particular nuclei
- 7.3 understand the terms atomic (proton) number, mass (nucleon) number and isotope
- 7.4 understand that alpha and beta particles and gamma rays are ionising radiations emitted from unstable nuclei in a random process
- 7.5 describe the nature of alpha and beta particles and gamma rays and recall that they may be distinguished in terms of penetrating power
- 7.6 describe the effects on the atomic and mass numbers of a nucleus of the emission of each of the three main types of radiation
- 7.7 understand how to complete balanced nuclear equations
- 7.8 understand that ionising radiations can be detected using a photographic film or a Geiger-Muller detector
- 7.9 explain the sources of background radiation
- 7.10 understand that the activity of a radioactive source decreases over a period of time and is measured in becquerels
- 7.11 understand the term 'half-life' and understand that it is different for different radioactive isotopes
- 7.12 use the concept of half-life to carry out simple calculations on activity
- 7.13 describe the uses of radioactivity in medical and non-medical tracers, in radiotherapy, and in the radioactive dating of archaeological specimens and rocks

- 7.14 describe the dangers of ionising radiations, including:
- radiation can cause mutations in living organisms
  - radiation can damage cells and tissue
  - the problems arising in the disposal of radioactive waste
- and describe how the associated risks can be reduced.

### **c) Particles**

*Students will be assessed on their ability to:*

- 7.15 describe the results of Geiger and Marsden's experiments with gold foil and alpha particles
- 7.16 describe Rutherford's nuclear model of the atom and how it accounts for the results of Geiger and Marsden's experiment and understand the factors (charge and speed) which affect the deflection of alpha particles by a nucleus
- 7.17 understand that a nucleus of U-235 can be split (the process of fission) by collision with a neutron, and that this process releases energy in the form of kinetic energy of the fission products
- 7.18 understand that the fission of U-235 produces two daughter nuclei and a small number of neutrons
- 7.19 understand that a chain reaction can be set up if the neutrons produced by one fission strike other U-235 nuclei
- 7.20 understand the role played by the control rods and moderator when the fission process is used as an energy source to generate electricity.